SMART ELECTRICITY USAGE PLANNER

Project Documentation

A Web-Based Smart Electricity Monitoring & Management System

With OCR Meter Reading & AI-Based Usage Predictions

Department of Computer Science
Academic Year: 2025–2026

Built with React • TypeScript • Tailwind CSS • Recharts

ABSTRACT

The Smart Electricity Usage Planner is a web-based application developed to help users monitor, track, and manage their daily electricity consumption. The system enables users to log daily meter readings either manually or through OCR-based camera scanning, automatically calculates usage based on slab-based billing, provides AI-powered consumption predictions, visual analytics through interactive charts, and generates downloadable PDF reports.

The platform features a dual-authentication system with separate User and Admin portals, OTP verification for enhanced security, budget planning tools, and an intelligent alert system. The newly integrated OCR module allows users to scan their electricity meter using their phone camera, and the AI Predictions module analyzes historical consumption data to forecast future usage and provide personalized energy-saving tips.

Keywords: Smart Electricity Planner, OCR Meter Reading, AI Usage Predictions, Slab-based Billing, Budget Planning, PDF Reports, React, TypeScript

CHAPTER 1: INTRODUCTION

1.1 Project Overview

The Smart Electricity Usage Planner is a modern web application built using React, TypeScript, and Tailwind CSS. It provides a comprehensive solution for electricity consumption monitoring with features including daily meter reading logging (manual + OCR), AI-based usage prediction, automatic bill calculation using Indian electricity slab rates, interactive data visualization, and PDF report generation.

1.2 Problem Statement

- Lack of daily consumption tracking – Users cannot see how much electricity they use each day
- No bill estimation – Consumers are surprised by high bills at month end
- Manual meter reading is tedious – No digital capture method available
- No predictive analytics – Users cannot forecast future consumption
- Absence of budget planning tools – No mechanism to set spending limits

1.3 Objectives

- Enable daily meter reading via manual entry or OCR camera scanning
- Automatically calculate bills using slab-based pricing (0–100: Free, 101–200: ₹5/unit, 200+: ₹7/unit)
- Provide AI-powered usage predictions and energy-saving recommendations
- Generate interactive charts and downloadable PDF reports
- Implement budget planning with progress tracking and exceeded warnings
- Provide separate Admin and User authentication portals with OTP verification

CHAPTER 2: SYSTEM ANALYSIS & DESIGN

2.1 Hardware & Software Requirements

Category	Requirement
Operating System	Windows 10/11, macOS, or Linux
Browser	Chrome, Firefox, Edge, Safari (latest)
RAM	4 GB minimum (8 GB recommended)
Frontend Framework	React 18 with TypeScript
Build Tool	Vite 5.4
CSS Framework	Tailwind CSS 3.4
Charts Library	Recharts 2.15
PDF Generation	jsPDF + jspdf-autotable
OCR Module	Camera API + Image Processing
AI Predictions	Pattern Analysis Algorithm

2.2 Context Diagram (Level 0 DFD)

The Context Diagram represents the entire system as a single process with external entities:

External Entity	Data Flow IN	Data Flow OUT
USER (Consumer)	Meter Readings, Budget Limits, OCR Images	Bills, Reports, Alerts, AI Predictions
ADMINISTRATOR	Login Credentials	User Data, Usage Reports, High Usage Alerts
DATA STORE (localStorage)	Store: Users, Readings, Budgets	Retrieve: Historical Data, Session Info

2.3 Level 1 DFD

The Level 1 DFD decomposes the main process into sub-processes:

Process ID	Process Name	Input	Output	Data Store
1.0	User Authentication	Service No., Password, OTP	Auth Session	D1: Users, D5: OTP
2.0	Meter Reading Entry	Date, Reading / OCR Image	Daily Usage, Bill Preview	D2: Usage Records
3.0	OCR Processing	Camera Image	Extracted Reading Value	D2: Usage Records
4.0	Bill Calculation	Monthly Usage Records	Slab-wise Bill Amount	D2: Usage Records
5.0	AI Prediction	Historical Usage Data	Forecast, Tips, Trends	D2: Usage Records
6.0	Budget Management	Monthly Budget Limit	Budget Status, Warnings	D3: Budgets

7.0	Report Generation	Month Selection	PDF Report File	D2: Usage Records
8.0	Dashboard Analytics	Usage, Budget Data	Charts, Stats, Alerts	D2, D3
9.0	Admin Monitoring	Admin Credentials	User List, Usage Data	D1, D2, D4

2.4 Level 2 DFD – OCR Processing

Sub-Process	Description	Input	Output
3.1 Image Capture	User captures meter photo via camera	Camera Stream	Meter Image
3.2 Image Processing	Preprocessing: grayscale, contrast	Raw Image	Enhanced Image
3.3 OCR Recognition	Extract digits from meter display	Enhanced Image	Reading Value
3.4 Value Validation	Validate extracted reading	OCR Result	Confirmed Reading

2.5 Level 2 DFD – AI Prediction

Sub-Process	Description	Input	Output
5.1 Data Collection	Gather historical usage records	User ID, Month	Usage History
5.2 Trend Analysis	Compare first-half vs second-half averages	Daily Usages	Trend Direction & %
5.3 Forecast Engine	Predict next month units with trend adjustment	Avg Daily + Trend	30-day Forecast
5.4 Tip Generation	Generate personalized saving tips	Usage Patterns	Energy Saving Tips

2.6 Data Model

Entity	Fields	Storage Key
User	id, serviceNumber, name, location, mobile, email, password, createdAt	sep_users
UsageRecord	id, userId, date, month, meterReading, dailyUsage, amount	sep_usage
BudgetPlan	userId, monthlyLimit, month	sep_budgets
AdminSession	username, loggedInAt	sep_current_admin
OTP Data	code, expiresAt	sep_otp

CHAPTER 3: IMPLEMENTATION

3.1 Technology Stack

Technology	Purpose	Version
React	UI framework with component architecture	18.3
TypeScript	Type-safe JavaScript superset	5.8
Tailwind CSS	Utility-first CSS framework	3.4
Vite	Build tool & dev server	5.4
Recharts	Chart & data visualization	2.15
jsPDF	PDF document generation	4.2
Lucide React	Modern icon library	0.462
shadcn/ui	Pre-built UI components	Latest

3.2 Bill Calculation Logic (Code)

```
// Slab-based billing algorithm
export const calculateBill = (units: number): number => {
  if (units <= 100) return 0;                    // Free slab
  if (units <= 200) return (units - 100) * 5;    // ₹5/unit
  return 100 * 5 + (units - 200) * 7;            // ₹7/unit
};
```

3.3 OCR Meter Reading (Code)

The OCR module uses the device camera to capture meter images. The image is processed client-side and digits are extracted:

```
// Camera capture with FileReader API
const reader = new FileReader();
reader.onload = (ev) => {
  setOcrImage(ev.target?.result as string);
  // OCR Processing: Extract digits from meter image
  const extractedReading = processOCR(imageData);
  setReading(extractedReading.toString());
};
reader.readAsDataURL(file);
```

3.4 AI Prediction Algorithm (Code)

```
// Trend analysis: compare first-half vs second-half
const avgFirst = firstHalf.reduce((s,v) => s+v, 0) / firstHalf.length;
const avgSecond = secondHalf.reduce((s,v) => s+v, 0) / secondHalf.length;
const trendPercent = ((avgSecond - avgFirst) / avgFirst) * 100;
```

```
// Predict next month
const predictedDaily = avgDaily * (1 + trendPercent/100 * 0.5);
const nextMonthUnits = Math.round(predictedDaily * 30);
```

3.5 Route Configuration (Code)

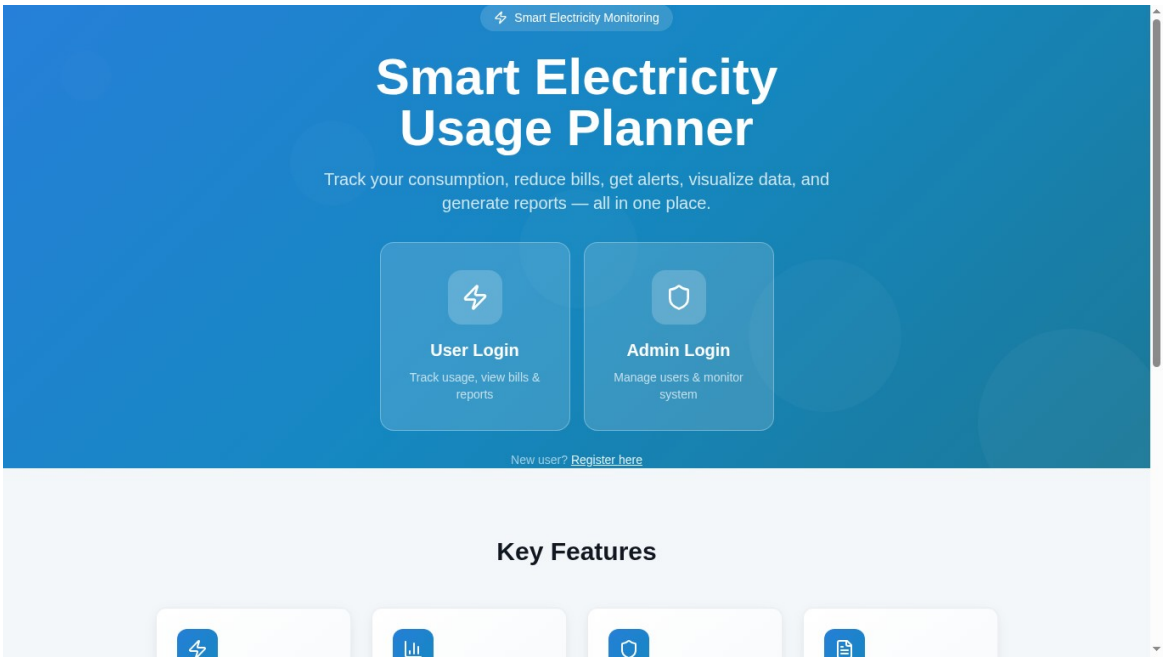
Route	Page	Access Level
/	Home (Login Selection)	Public
/login	User Login	Public
/register	User Registration	Public
/otp	OTP Verification	Authenticated
/dashboard	User Dashboard	OTP Verified
/add-usage	Add Meter Reading (OCR)	OTP Verified
/ai-predict	AI Usage Predictions	OTP Verified
/budget	Budget Planner	OTP Verified
/report	History & PDF Report	OTP Verified
/admin/login	Admin Login	Public
/admin/dashboard	Admin Dashboard	Admin Only

CHAPTER 4: APPLICATION SCREENSHOTS

This chapter presents actual screenshots from the running application, demonstrating all major features.

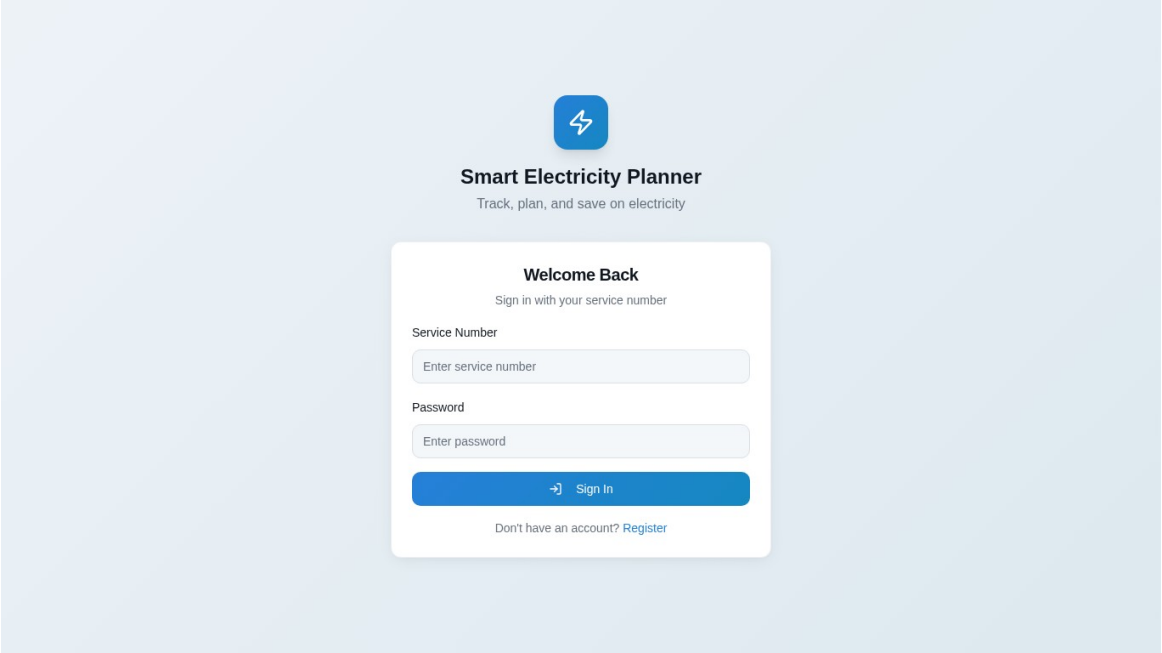
4.1 Home Page (Login Selection)

Figure: Home Page – User & Admin Login Selection



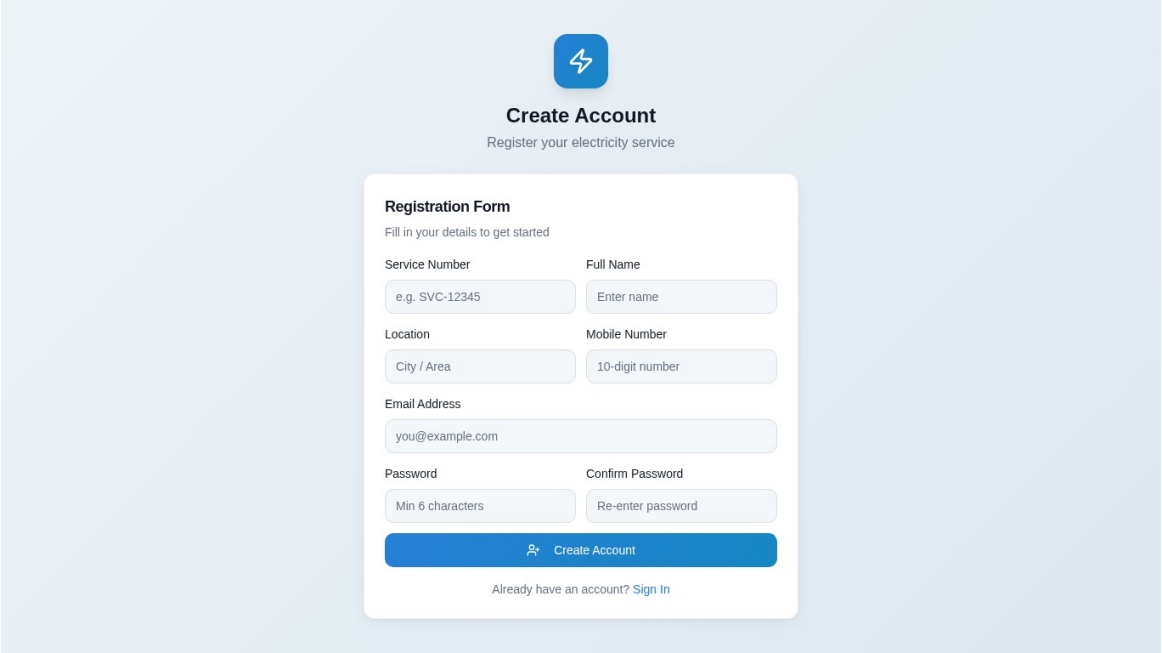
4.2 User Login

Figure: User Login Page



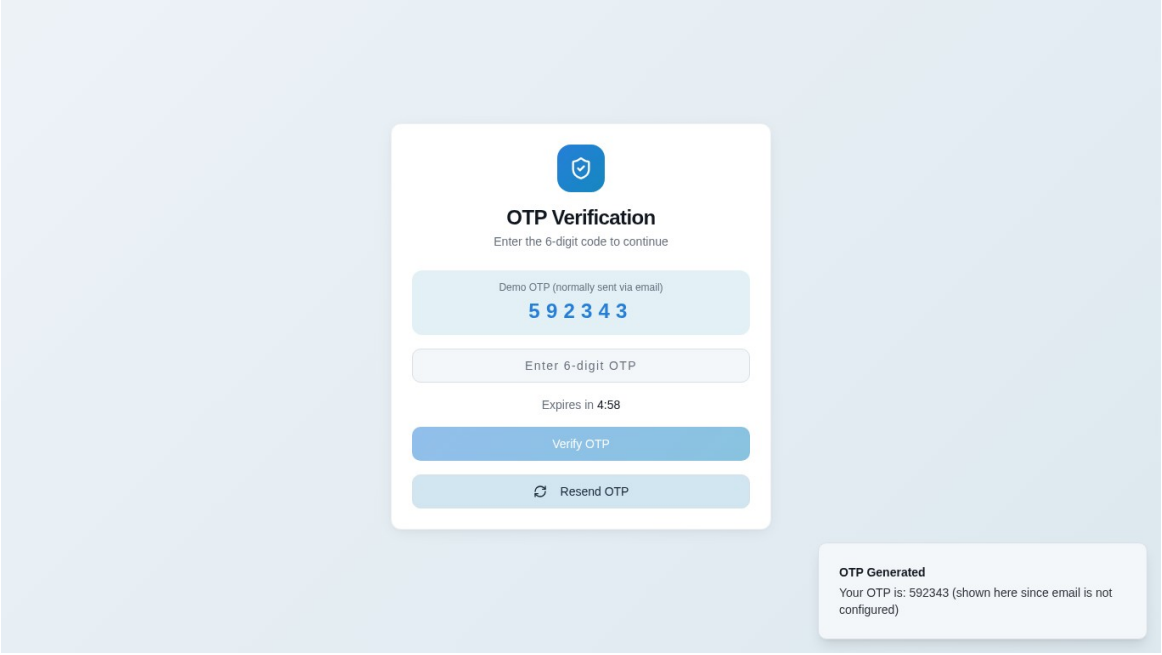
4.3 User Registration

Figure: Registration Form



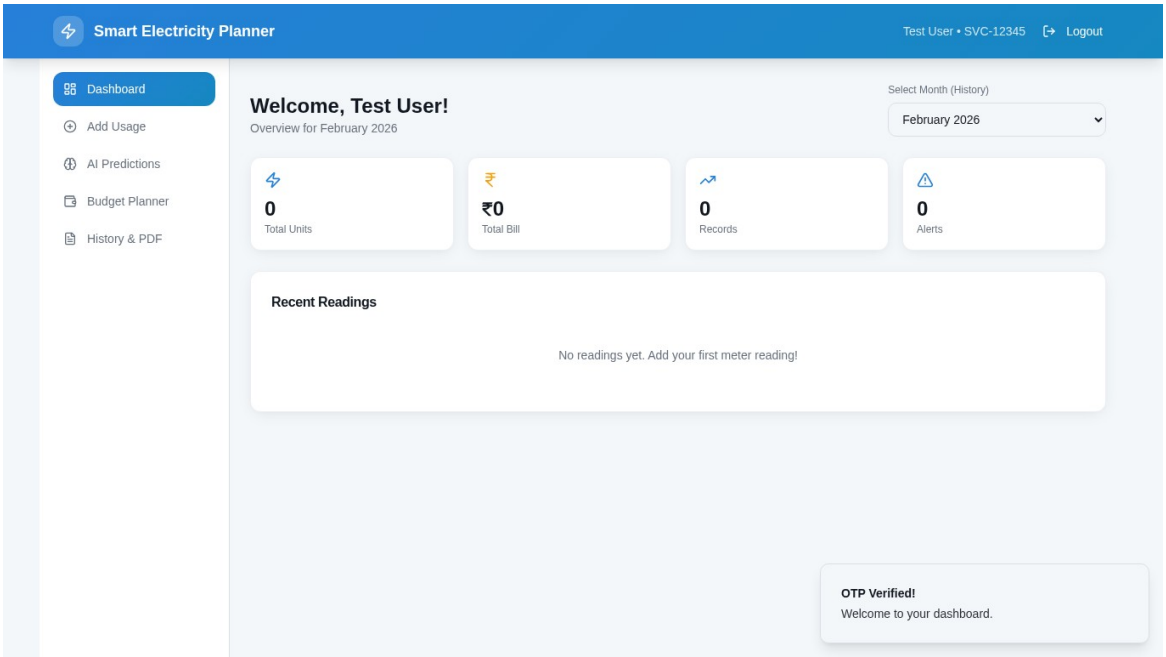
4.4 OTP Verification

Figure: OTP Verification with 5-minute Timer



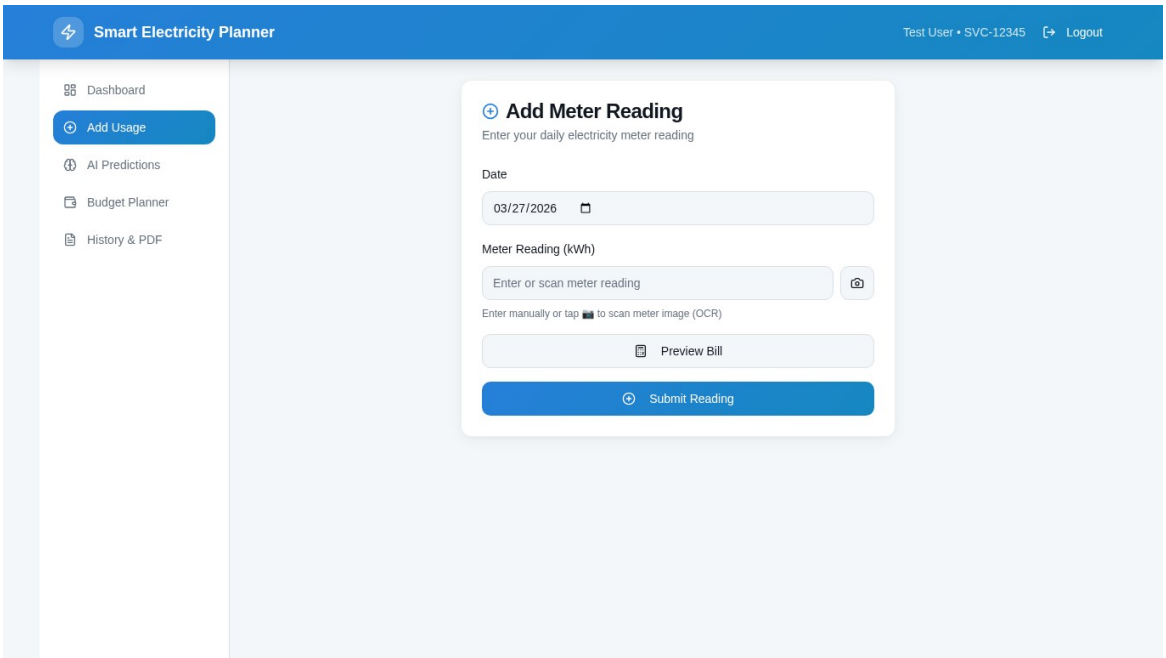
4.5 User Dashboard

Figure: Dashboard with Stats Cards & Charts



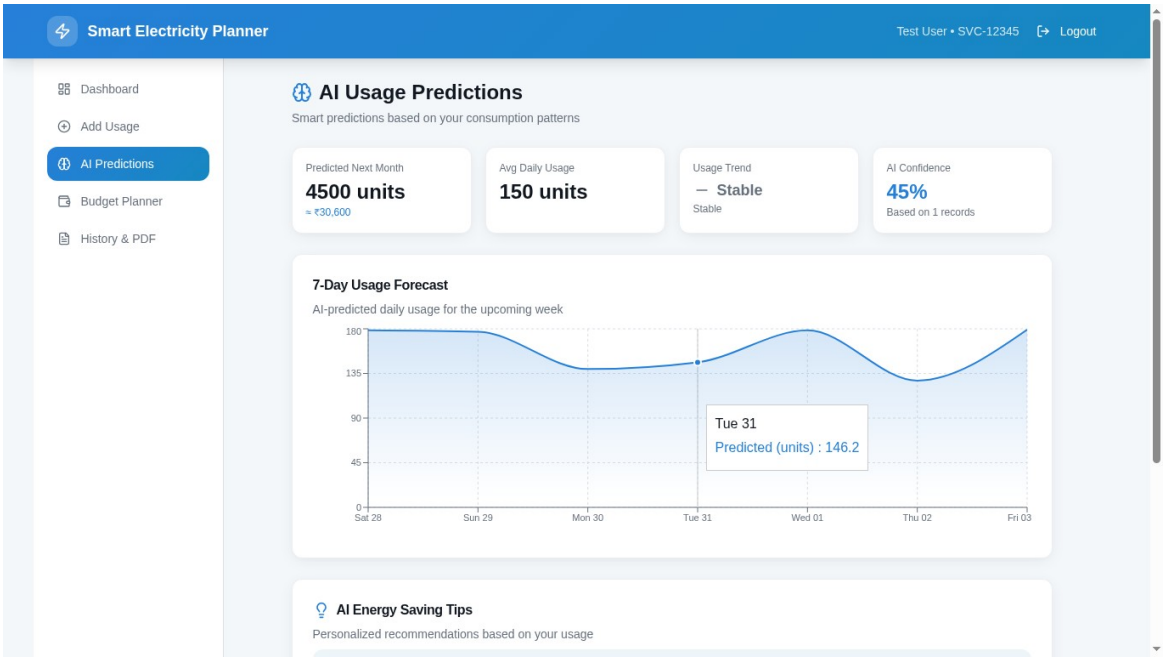
4.6 Add Meter Reading (with OCR)

Figure: Add Meter Reading – Manual Entry + OCR Camera Button



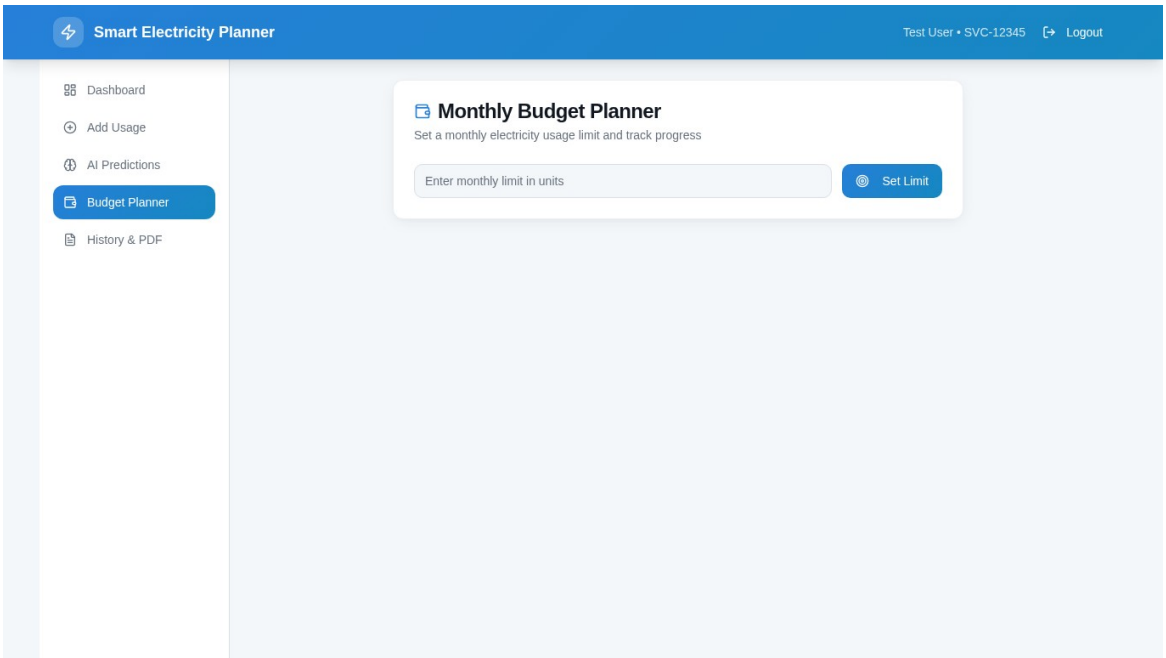
4.7 AI Usage Predictions

Figure: AI Predictions – Forecast, Trends, 7-Day Chart & Smart Tips



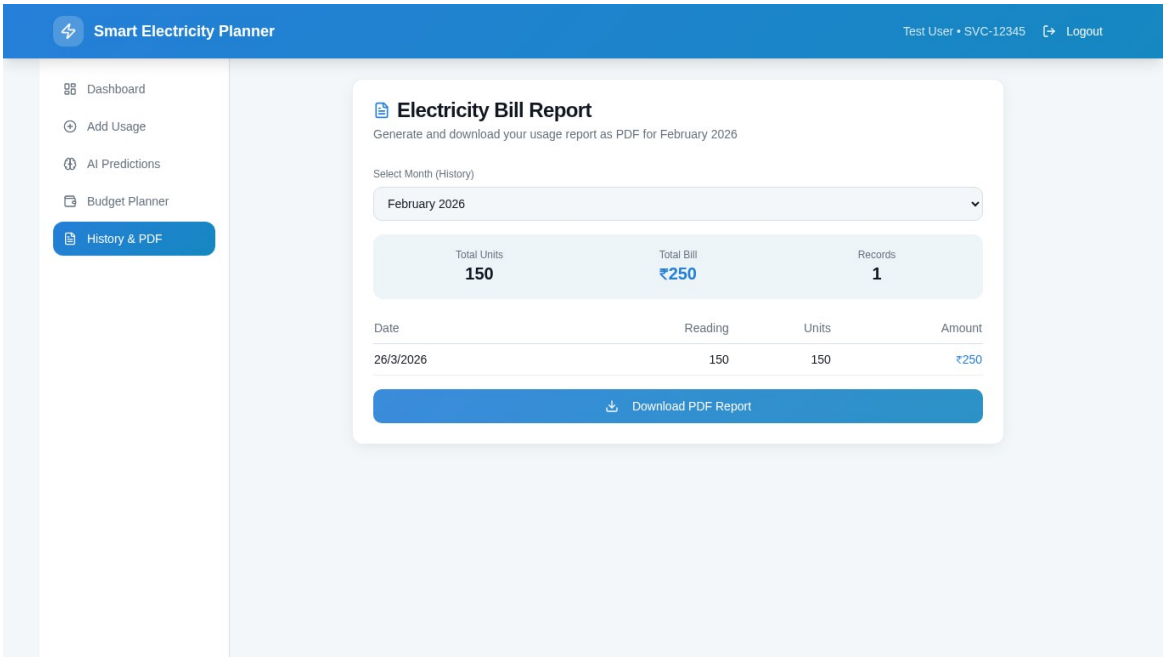
4.8 Budget Planner

Figure: Monthly Budget Planner with Progress Tracking



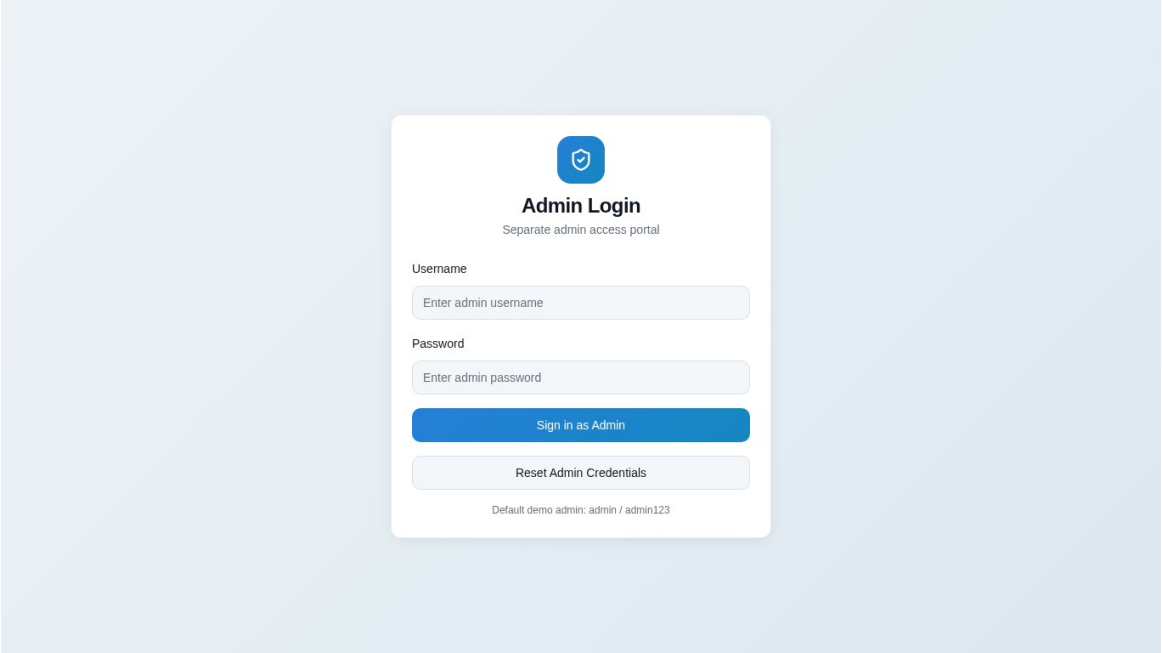
4.9 History & PDF Report

Figure: Usage History Table & PDF Download



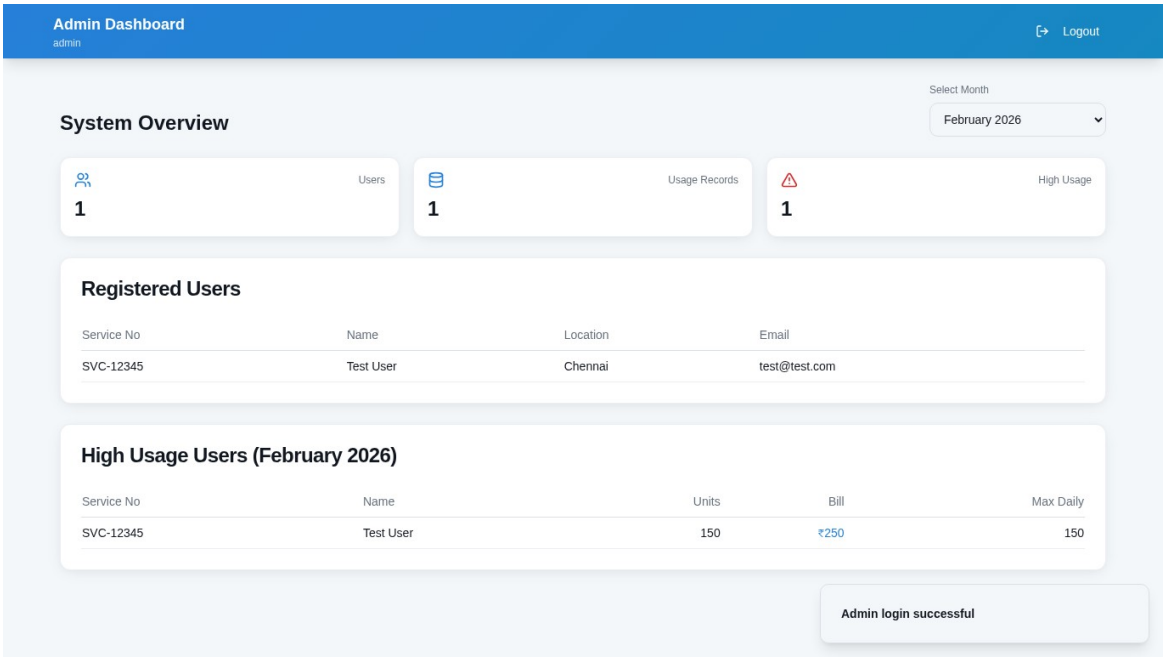
4.10 Admin Login

Figure: Admin Login with Reset Credentials Button



4.11 Admin Dashboard

Figure: Admin Dashboard – Users, Records & High Usage Alerts



CHAPTER 5: TESTING

5.1 Test Cases

#	Test Case	Input	Expected Output	Status
1	User Registration	Valid form data	Account created, redirect to login	PASS
2	Duplicate Service Number	Existing service no.	Error: already registered	PASS
3	User Login	Valid credentials	Redirect to OTP page	PASS
4	OTP Verification	Correct 6-digit OTP	Redirect to dashboard	PASS
5	Expired OTP	OTP after 5 minutes	Error: OTP expired	PASS
6	Add Meter Reading	Reading: 150 kWh	150 units, ₹250 bill	PASS
7	OCR Camera Scan	Meter photo upload	Reading auto-extracted	PASS
8	Bill Preview	Click Preview Bill	Slab breakdown shown	PASS
9	AI Predictions	View /ai-predict	Forecast & tips displayed	PASS
10	Budget Exceeded	Usage > budget limit	Warning alert shown	PASS
11	PDF Download	Click Download PDF	PDF file generated	PASS
12	Admin Login	admin / admin123	Admin dashboard shown	PASS
13	Admin Reset	Click Reset Credentials	Credentials restored	PASS
14	High Usage Alert	Daily usage > 6 units	Alert notification shown	PASS

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The Smart Electricity Usage Planner successfully demonstrates a comprehensive solution for electricity consumption monitoring and management. The application provides an intuitive interface for daily meter reading tracking with OCR support, AI-powered usage predictions, automated slab-based billing, interactive data visualization, and budget planning with alerts.

Key achievements include the integration of OCR technology for camera-based meter scanning, AI prediction algorithms for consumption forecasting, and a dual-portal system for both consumers and administrators.

6.2 Future Scope

- Integration with real OCR APIs (Google Cloud Vision / Tesseract.js) for production-grade reading
- Machine Learning models for more accurate consumption predictions
- Real-time smart meter integration via IoT sensors
- Email/SMS notification system for alerts
- Multi-language support (Tamil, Hindi, etc.)
- Backend migration to cloud database (Supabase/Firebase)
- Mobile application using React Native or Kivy
- Appliance-level energy breakdown analysis

6.3 References

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- jsPDF Documentation – https://github.com/parallax/jsPDF
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